

CYBERSECURITY SECURITY ASSESSMENT

Report 5

SSH Authentication Event Investigation

Laboratory-based security assessment using Kali Linux and Metasploitable 2

Prepared as a practical cybersecurity assessment report

1. Introduction

Secure Shell (SSH) is a network protocol commonly used for remote administration of Linux and Unix-based systems. Because SSH provides access to system resources through authenticated sessions, its authentication events are important from a security monitoring and incident-analysis perspective. Failed authentication attempts can indicate incorrect credentials, unauthorized access attempts, or other suspicious activity, while successful authentication events can confirm that access was eventually established. This practical assessment investigates SSH authentication activity in an isolated laboratory environment. The assessment uses a Kali Linux system as the source system and a Metasploitable 2 virtual machine as the target. The objective is not to claim a confirmed brute-force attack from limited evidence, but to demonstrate how an analyst can identify an exposed SSH service, observe authentication failures, verify a successful login, and correlate those events with the target system's authentication logs.

2. Objective

The objective of this assessment was to identify the SSH service exposed by the target, perform controlled authentication tests in the laboratory environment, inspect the resulting authentication events, and correlate the network activity with server-side log entries. The assessment also demonstrates how a failed authentication event and a later successful authentication event can be examined as part of a basic security investigation.

3. Laboratory Environment

The assessment was performed inside a controlled virtualized environment. Kali Linux was used as the assessment and analysis system, while Metasploitable 2 was used as the intentionally vulnerable target. Both systems communicated through the laboratory virtual network.

Component	Configuration / Role
Assessment system	Kali Linux
Target system	Metasploitable 2
Target IP address	192.168.202.130
Observed service	SSH — TCP port 22
Primary tools	Nmap, SSH, Linux authentication logs
Environment	Isolated virtual laboratory

4. Methodology

The assessment began by identifying the network services available on the Metasploitable target. Nmap was used to perform a service discovery scan and confirm whether TCP port 22 was open. After confirming SSH availability, a controlled authentication test was performed from Kali Linux. An unsuccessful authentication attempt was then verified from the server-side authentication log. A valid laboratory login was subsequently performed using the supplied Metasploitable credentials. After authentication, the identity of the logged-in account and the target hostname were verified using standard Linux commands. The server-side authentication log was then examined again to locate the corresponding successful authentication event. The failed and successful events were compared to understand the sequence of activity.

5. SSH Service Identification

The first stage was service discovery. Nmap was used against the target IP address to identify listening network services. The scan reported TCP port 22 as open and identified the service as SSH. This established that the target exposed a remote administration service that could subsequently be investigated through controlled authentication testing.

```
sudo nmap 192.168.202.130
```

```
(kali@kali)-[~]
$ sudo nmap 192.168.202.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-30 06:41 -0400
Nmap scan report for 192.168.202.130
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds

(kali@kali)-[~]
$ sudo nmap -p 22 192.168.202.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-30 06:42 -0400
Nmap scan report for 192.168.202.130
Host is up (0.00072s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds

(kali@kali)-[~]
$
```

Nmap service-discovery result showing 22/tcp open and identified as SSH.

The result provides the initial technical basis for the investigation. The presence of an open SSH service does not by itself prove that the service is vulnerable; it establishes the attack surface that must be examined further.

6. Failed SSH Authentication Event

A controlled SSH authentication attempt was made using an incorrect password. The purpose of this step was to generate a failed authentication event that could be verified directly on the target system. This demonstrates an important principle of security monitoring: authentication activity can often be observed both from the client side and from server-side logs.

```
ssh msfadmin@192.168.202.130
```



Metasploitable authentication log showing a failed password event for the msfadmin account.

The log entry records the failed password attempt and identifies the source address. This provides stronger evidence than the SSH client prompt alone because it confirms that the target system itself recorded the authentication failure.

7. Successful SSH Authentication

A subsequent controlled login was performed with the valid laboratory credentials. The session was established successfully, demonstrating that the SSH service accepted the credentials and created an authenticated shell session.

```
ssh -oHostKeyAlgorithms=+ssh-rsa msfadmin@192.168.202.130
```

```
(kali@kali)-[~]  
$ ssh -oHostKeyAlgorithms=+ssh-rsa msfadmin@192.168.202.130  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
msfadmin@192.168.202.130's password:  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
Last login: Sun Aug 30 06:18:48 2026  
msfadmin@metasploitable:~$ whoami  
msfadmin  
msfadmin@metasploitable:~$ hostname  
metasploitable  
msfadmin@metasploitable:~$
```

Successful SSH login to Metasploitable 2 followed by verification of the active account and hostname.

After authentication, the account identity and hostname were verified. The output confirms that the session was operating as the msfadmin account on the metasploitable host.

whoami

hostname

8. Server-Side Verification of Successful Authentication

The successful login was then verified from the target's authentication log. The log contains an 'Accepted password' entry associated with the msfadmin account and the source address of the Kali system. This server-side evidence confirms that the successful SSH session was recorded by the target.

grep "Accepted password" /var/log/auth.log

```
msfadmin@metasploitable:~$ grep "Accepted password" /var/log/auth.log
Aug 30 07:01:19 metasploitable sshd[8099]: Accepted password for msfadmin from 1
92.168.202.131 port 59966 ssh2
msfadmin@metasploitable:~$
```

Authentication log showing the accepted SSH password event and source address.

The combination of the successful SSH session and the corresponding 'Accepted password' log entry establishes a direct relationship between the authentication action and the server-side record.

9. Authentication Event Correlation

To examine the activity as an investigation rather than as isolated commands, failed and accepted authentication records were correlated from the same authentication log. The resulting evidence shows a failed password event followed later by an accepted password event for the same account and source system.

```
grep -E "Failed password|Accepted password" /var/log/auth.log
```



Correlated authentication log entries showing failed and accepted SSH authentication events.

This timeline demonstrates the value of log correlation. A single failed login is not sufficient evidence to classify an incident as a brute-force attack. However, correlating failed and successful events provides a clear record of authentication activity and allows an analyst to investigate whether the sequence is expected or suspicious.

10. Findings and Security Analysis

The assessment confirmed that the target exposed SSH on TCP port 22 and that the service accepted remote authentication. A failed password attempt was successfully recorded in the target's authentication log, followed later by a successful authentication using valid credentials. The source address recorded by the target provides an additional correlation point between the events. The practical significance of these results is that SSH authentication should not be evaluated only from the perspective of whether a login succeeds. Authentication logs provide visibility into failed and successful attempts and can be used to construct a basic event timeline. In a production environment, repeated failures, unusual source addresses, successful logins following multiple failures, unusual login times, and access to sensitive systems would warrant additional analysis. The evidence collected in this laboratory does not, by itself, establish a confirmed brute-force attack. The correct finding is that SSH authentication activity was observed and successfully correlated through client-side actions and server-side authentication logs.

11. Recommendations

In a real production environment, SSH exposure should be limited to trusted administrative networks or hosts wherever possible. Strong authentication should be enforced, with key-based authentication preferred over password-only access. Accounts should use strong, unique credentials and unnecessary accounts should be disabled. Authentication logs should be centrally collected and monitored so that repeated failures and unusual successful logins can be detected. Rate limiting or controls such as fail2ban may also be used to reduce repeated authentication attempts. Administrative access should additionally be protected through appropriate network segmentation, firewall rules, and least-privilege principles. For incident response, security teams should correlate SSH authentication events with source IP addresses, timestamps, user accounts, system activity, and other available logs before determining whether an event represents malicious activity.

12. Conclusion

This assessment demonstrated a complete, controlled workflow for investigating SSH authentication activity on a vulnerable laboratory system. The process began with network reconnaissance, where Nmap identified TCP port 22 as an open SSH service on the Metasploitable target. This established the service under investigation and provided the basis for the subsequent authentication tests. A failed SSH authentication attempt was then generated and verified through the target's authentication log. A valid authentication was subsequently performed, and the resulting session was confirmed using the `whoami` and `hostname` commands. The successful authentication was independently verified through an 'Accepted password' entry in `/var/log/auth.log`. Finally, the failed and successful events were correlated to create a basic authentication timeline. The main security lesson from this exercise is that authentication monitoring depends on evidence from multiple sources. An SSH prompt can show whether a login succeeded or failed, but server-side logs provide the information required to investigate when the event occurred, which account was targeted, and where the connection originated. In a real security operation, this same approach can be expanded by correlating authentication records with firewall logs, endpoint telemetry, SIEM alerts, user activity, and other network evidence. The laboratory evidence therefore demonstrates practical familiarity with SSH service identification, controlled authentication testing, Linux authentication logs, event correlation, and basic security analysis. It also demonstrates the importance of avoiding unsupported conclusions: the observed events show authentication activity, but additional evidence would be required before classifying the activity as a confirmed brute-force or compromise event.

13. References

Nmap Documentation — Network discovery and service scanning reference.

OpenSSH Documentation — SSH protocol and authentication concepts.

Metasploitable 2 — Intentionally vulnerable Linux laboratory environment.

Ubuntu Server Documentation — Linux system administration and SSH configuration.

Linux Manual Pages (man pages) — Documentation for commands such as ssh, whoami, hostname, and grep.

OpenSSH Server Manual (sshd) — SSH server configuration and authentication behaviour.

Linux /var/log/auth.log — Authentication event logging used for laboratory verification.

MITRE ATT&CK — Remote Services: SSH (T1021.004) — Reference for understanding SSH as a remote-access technique.

OWASP Logging Cheat Sheet — Security logging, monitoring and event analysis practices.

NIST Computer Security Incident Handling Guide (SP 800-61) — General principles for identifying, analysing and responding to security incidents.